MIL-L-44033A
25 October 1984
SUPERSEDING
MIL-L-44033
3 October 1980

MILITARY SPECIFICATION

LINES, MULTI-LOOP, CONCENTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of multi-loop lines fabricated of webbing and intended for use in airdrop operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	-	Twine, Fibrous Cotton, Wrapping
UU-P-268	-	Paper, Kraft, Wrapping
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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MILITARY

MIL-L-10547 - Liner, Case, and Sheet, Overwrap Water-Vaporproof or Waterproof, Flexible

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

68F217 - Line-Multi-Loop, Concentric Loop Type, Assembly of

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.3 and 6.6).

3.2 Materials. Materials shall conform to Drawing 68F217, applicable documents and requirements specified herein.

3.2.1 Webbing. The color shade of webbing specified on Drawing 68F217 shall match the standard shade sample (see 6.4).

3.2.2 Wax. The wax used on the sliding keeper and the ends of nylon webbing used as buffers shall be a mixture consisting of 50 percent each of paraffin and beeswax (see 4.4.1).

3.2.3 Marking medium. The contractor shall certify that the medium used for stenciling the identification of the slings or extraction lines shall be as specified on Drawing 68F217.

3.3 Construction. Construction shall conform to Drawing 68F217 and requirements specified herein.

3.3.1 Webbing relaxation. All webbing used in the fabrication of loops shall be relaxed for a minimum of 48 hours prior to measuring for cutting. Relaxation of the webbing shall be accomplished by loosening the coiled rings so there is a minimum of 1/4 inch space between the webbing coils of the roll.

3.3.2 Dimensions. Finished lengths and the loops of the lines shall be as specified in the contract (see 6.3). The cut length of the individual loops, after relaxation of the webbing rolls (see 3.3.1) and prior to splicing, shall be determined by the contractor and shall result in the finished length shown on Drawing 68F217. Concentric loops shall fit one within the other so that when inspected as specified in 4.4.4, variations in the lengths of the loops shall be as specified on Drawing 68F217.

3.3.3 Machine stitching. All machine stitching shall conform to FED-STD-751 and the type specified on the applicable drawings. Ends of stitching shall be back stitched not less than 1/2 inch. Thread tension shall be maintained so there will be no loose stitching and the lock shall be embedded into the materials sewn. All thread ends shall be trimmed so that they do not exceed 1/4 inch.

3.3.3.1 Automatic stitching. Automatic sewing machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.3.3.2 Loose stitches. Loose stitches shall not be removed but shall be reinforced by adjacent stitching.

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3.3.3.3 Tight stitches. Tight stitches which occur on the sliding keeper only shall be removed and the section shall be restitched.

3.3.3.4 Repairs of the 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, the stitching shall be repaired by adjacent stitching with the stitches extending a minimum of 1 inch beyond the ends of the stitches being repaired. 1/

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by adjacent stitching. The stitching shall start a minimum of 1 inch in back of the defective area, and continue a minimum of 1 inch beyond the ends of the stitching being repaired. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

3.3.4 Wax dipping. All ends of nylon webbing used for the buffers, shall be dipped not less than 1 inch into the wax mixture specified in 3.2.2. The sliding keepers shall be stitched and shaped to the contour of the assembled line, after which they shall be completely dipped in wax. During the dipping operation the wax temperature shall be maintained at 180° ± 20°F (see 4.4.2).

3.3.5 Searing. Except for the buffers all ends of nylon webbing shall be seared to prevent fraying. Ends shall be smooth, even, and free of sharp edges.

3.4 Identification marking. Using the marking medium specified on drawing 68 F217, all marking shall be of characters 1/2 inches in height. Markings shall be located near each end of the line on one side, and shall be placed so that they will not be obscured by the keepers. If necessary, both sides of the 3 foot line may be marked. No marking shall appear on slices or keepers. Load capacity shall not be included in the identification information. Identification shall be as follows:

a. US
b. Part No.
c. Finished length
d. Date of manufacture
e. Manufacturer's name or code

3.5 Workmanship. The finished item shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (See 6.3), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. Components and materials listed below shall be accepted on the basis of a contractor's certificate of compliance with the indicated requirements.

Component	Characteristic	Requirement paragraph
Marking medium	Specified on Drawing 68F217	3.2.3
Wax	Composition	3.2.2

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether construction details which cannot be examined in the finished product are in accordance with specified requirements. Whenever nonconformance is noted, corrections shall be made to the items affected and lot in-process. Items which cannot be corrected shall be removed from production.

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4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of lines of one length, loop number, and webbing type only. The sample unit shall be one line. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects for lines up to 60 feet long and 4.0 for major defects and 10.0 for total (major and minor combined) defects for lines more than 60 feet long.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Webbing	Not specified type or class.	X	
	Color not as specified.		X
	Ends not wax dipped or seamed as specified.		X
	Hole, tear, or cut in material.	X	
	Abrasion resulting in rupture of yarns, or in nap sufficient to obscure the identity of any yarns exceeding 10 percent of width or 1 inch in length	X	
	Edge frayed, beaded, corded, or torn.	X	
	Two broken warp or fill yarns.		X
	More than two broken warp or fill yarns.	X	
Open seams	1/2 inch or less.		X
	More than 1/2 inch.	X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.		
Run-off (see open seams)			
Stitch type	Wrong stitch type.	X	
Stitch tension	Loose resulting in loose top or bobbin thread.		X
	Excessively tight resulting in puckering of webbing (applicable to sliding keepers only).		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Stitches per inch	One stitch per inch more than maximum specified.		X
	More than one stitch per inch more than maximum specified.	X	
	One stitch per inch less than minimum specified.		X
	More than one stitch per inch less than minimum specified.	X	
	NOTE: Variation in the number of stitches per inch caused by the operator speeding up machine to sew over heavy places, shall be classified as follows: (a) Within the major defect classification minor defect; (b) Within the minor defect classification - no defect.		
Thread breaks, skipped stitches, or runoffs	Not adjacent stitched as specified.		X
	NOTE: Thread breaks or two or more consecutive skipped or runoff stitches not adjacent stitched shall be classified as open seams.		
Ends of stitching	Not secured as specified.		X
Rows of stitching	Any row of stitching missing, except on 72 continuous lines and six point stitching patterns.	X	
	On 72 continuous lines of stitching one row more or less than specified.	X	
	On six point stitch pattern, more or less than one row specified per individual stitch pattern.	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Components and assembly	Any component part omitted or not as specified or required operation omitted (unless otherwise classified herein).	X	
	Any component misplaced or not assembled as specified.	X	
	Needle chew.		X
	Any darn, splice or other unauthorized repair.	X	
Sliding keeper	Not specified size or type.	X	
	Sliding keeper not completely dipped in melted wax mixture.		X
	Stitch pattern on overlap of keepers not in location shown on Drawing 68F217 in relation to top or bottom loops.		X
	One or both sliding keepers omitted.	X	
Fixed keeper	Not specified size or type.	X	
	Not assembled as specified.		X
	One or more fixed keepers omitted.	X	
Splices	Not spaced alternately as specified.		X
	Not joined parallel as specified.	X	
	More than two splices per loop.	X	
Buffer	Not specified size or type.	X	
	Not assembled as specified.		X
	One or both buffers omitted.	X	
	Not longitudinally stitched to loop.	X	
Cleanness	Dirt, grease, oil or foreign matter clearly noticeable.		X
Identification marking	Missing, incomplete, not legible, improper location or size.		X
	Wrong color.		X

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4.4.4 End item dimensional examination. The end item shall be examined for conformance to the specified dimensions on the applicable drawing. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of lines of one length, loop number, and webbing size only. The sample unit shall be one line. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed in table II shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE II. Packaging defects

Category	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Any component missing, damaged, defective, or not of specified material.
Workmanship	Inadequate coiling, folding, or tying of the end item. Bulged or distorted container. Incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling or contents loose or not snug-fitting.
Contents	Number is more or less than required.

4.4.5.1 Palletization examination. An examination shall be made to determine that the palletization complies with section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Loads not bonded with required straps as specified. Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Weight	Exceeds maximum load limits.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.3).

5.1.1 Level A. Each multi-loop line shall be coiled or lap folded and securely tied with twine conforming to type I or II, 8 ply of T-T-871.

5.1.2 Commercial. Each multi-loop line shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.3).

5.2.1 Level A packing. Lines of one length and number of loops only, preserved as specified in 5.1, shall be packed in quantities as specified in table III in a snug-fitting shipping container conforming to overseas type, style A or J, type 2 load of PPP-B-601. Each shipping container shall be waterproofed with a sealed case liner conforming to type I, or II, grade C of MIL-L-10547; and closed and reinforced with steel strapping in accordance with the appendix of PPP-B-601.

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TABLE III. Quantity of lines per container

Lengths (ft)	1 Loop	2 Loop	3 Loop	4 Loop
3	100	50	35	25
9	40	20	10	10
11	35	15	10	6
12	30	15	10	6
14	30	15	10	6
16	20	10	6	6
20	20	10	6	4
26	15	8	5	3
28	15	8	5	3
36	10	6	4	2
40	10	6	4	2
60	6	3	2	1

All lines longer than 60 feet will be packed one (1) per container.

5.2.2 Level B packing. Lines of one length and number of loops only, preserved as specified in 5.1, shall be packed in quantities as specified in table II in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant containers. When specified (see 6.3), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Lines, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Palletization. When specified (see 6.3), lines, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Each unit pack shall first have attached a manila colored paper shipping tag for the identification information.

6. NOTES

6.1 Intended use. The items are intended for use in the rigging, extraction, and delivery of cargo from aircraft in flight.

6.2 Assembly. As an aid in assembling formed loops into slings or lines, segments may be marked for identification at time of measuring, as No. 1 (inside loop), 2, 3 or 4 as indicated on drawings.

6.3 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Stock number.
c. Part number, finished length, number of loops and nomenclature (see 3.3.2).
d. When a first article is required (see 3.1, 4.3, and 6.6).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.4 Shade samples. For access to standard shade samples of webbing, address the contracting activity issuing the invitation for bids.

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6.5 International standardization agreement. Design provisions of this document are the subject of international standardization agreements NATO STANAG No. 3469 (Edition No. 2) and ASCC AIR STD 44/15E. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconcilation action through international standardization channels including Departmental Standardization Offices, if required.

6.6 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article should consist of one line of one length, loop number, and webbing type. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - AS
Air Force - 11

Preparing activity:

Army - GL

Project No. 1670-0691

Review activities:

Army - AV
Navy - MC
Air Force - 82, 99

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-44033A	Lines, Multi-Loop, Concentric

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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